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Treating lymphomas

Abstract

This document provides methods and materials related to treating lymphomas. For example, methods and materials relating to the use of a composition containing albumin-containing nanoparticle/antibody complexes (e.g., ABRAXANE®/anti-CD20 polypeptide antibody complexes) to treat lymphomas are provided.

Inventors: Markovic; Svetomir N. (Rochester, MN), Nevala; Wendy K. (Rochester, MN)

Applicant: Mayo Foundation for Medical Education and Research (Rochester, MN)

Family ID: 54834378

Assignee: Mayo Foundation for Medical Education and Research (Rochester, MN)

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 15/030,567, filed Apr. 19, 2016, which is a national stage application of PCT/US2015/035505, filed Jun. 12, 2015, which claims the benefit of U.S. Provisional Ser. No. 62/012,190, filed Jun. 12, 2014. The disclosures of the prior applications are considered part of (and are incorporated by reference in) the disclosure of this application.

BACKGROUND

1. Technical Field

(1) This document relates to methods and materials involved in treating lymphomas (e.g., mature B-cell neoplasms, mature T cell neoplasms, mature natural killer cell neoplasms, immunodeficiency-associated lymphoproliferative disorders, Hodgkin lymphomas, and non-Hodgkin lymphomas). For example, this document relates to methods and materials involved in using complexes containing albumin-containing nanoparticles (e.g., ABRAXANE® nanoparticles) and antibodies (e.g., anti-CD20 polypeptide antibodies such as Rituximab) to treat lymphomas.

2. Background Information

(2) Lymphomas are blood cancers that occur when B or T lymphocytes become cancerous. They can develop in the lymph nodes, spleen, bone marrow, blood, or other organs and eventually form a tumor.

SUMMARY

(3) This document provides methods and materials involved in treating lymphomas (e.g., mature B-cell neoplasms, mature T cell neoplasms, mature natural killer cell neoplasms, immunodeficiency-associated lymphoproliferative disorders, Hodgkin lymphomas, and non-Hodgkin lymphomas). For example, this document provides methods and materials for using complexes containing albumin-containing nanoparticles (e.g., ABRAXANE® nanoparticles) and antibodies (e.g., anti-CD20 polypeptide antibodies such as Rituximab) to treat lymphomas.

(4) ABRAXANE® is available from Celgene Corp. and is a nanoparticle formulation that combines paclitaxel with human albumin. Anti-CD20 polypeptide antibodies such as Rituximab are available from Genentech Inc., Roche, and Aryogen Biopharma under trade names such as Rituxan™, Mab Thera™, and Zytux™. RITUXAN®, MABTHERA™, and ZYTUX™ Rituximab is a chimeric monoclonal antibody against CD20 polypeptides present [s] on the surface of lymphocyte cells (see, e.g., U.S. Pat. No. 5,736,137).

(5) As described herein, in vitro mixing of albumin-containing nanoparticles (e.g., ABRAXANE® nanoparticles) and antibodies (e.g., Rituximab) can result in the formation of macromolecular complexes, the characteristics of which (e.g., size, antibody content, or chemotherapeutic drug content) can be customized depending on need. In some cases, such macromolecular complexes can retain antibody mediated target binding specificity, can retain or exhibit enhanced chemotherapeutic tumor cell cytotoxicity, and can exhibit no additional toxicity beyond that of ABRAXANE® nanoparticles alone. As also described herein, contacting ABRAXANE® with an anti-CD20 polypeptide antibody (e.g., Rituximab) prior to administration to a human (e.g., a human lymphoma cancer patient) can result in a complex that, when administered as a complex, has an increased ability to treat lymphoma as compared to a treatment regimen that includes administering ABRAXANE™ and the anti-CD20 polypeptide antibody separately in a manner that does not form ABRAXANE®/anti-CD20 polypeptide antibody complexes.

(6) The methods and materials provided herein can be used to increase the progression-free survival rate in lymphoma patients. Increasing progression-free survival can allow lymphoma

cancer patients to live longer.

(7) In general, one aspect of this document features a method for treating a mammal having lymphoma. The method comprises, or consisting essentially of, administering to the mammal a composition comprising nanoparticles containing albumin and paclitaxel complexed with an anti-CD20 polypeptide antibody under conditions wherein the length of progression-free survival is increased. The mammal can be a human. The lymphoma can be a Hodgkin lymphomas. The composition can comprise Rituximab complexed with the nanoparticles. The composition can comprise an alkylating agent complexed with the nanoparticles. The alkylating agent can be a platinum compound. The platinum compound can be carboplatin. The anti-CD20 polypeptide antibody can be a humanized antibody. The anti-CD20 polypeptide antibody can be a chimeric antibody. The composition can be administered by injection. The progression-free survival can be increased by 15 percent. The progression-free survival can be increased by 25 percent. The progression-free survival can be increased by 50 percent. The progression-free survival can be increased by 75 percent. The progression-free survival can be increased by 100 percent. The composition can be administered under conditions wherein the time to progression is increased.

(8) In another aspect, this document features a method for treating a mammal having lymphoma. The method comprises, or consists essentially of, administering, to the mammal, a composition comprising albumin-containing nanoparticle/antibody complexes, wherein the average diameter of the complexes is between 0.1 and 0.9 μm , and wherein the antibody is an anti-CD20 antibody. The mammal can be a human. The lymphoma can be a mature B-cell neoplasm. The lymphoma can be a mature T cell neoplasm. The lymphoma can be a Hodgkin lymphoma. The albumin-containing nanoparticle/antibody complexes can be ABRAXANE®/Rituximab complexes. The composition or the albumin-containing nanoparticle/antibody complexes can comprise an alkylating agent. The alkylating agent can be a platinum compound. The platinum compound can be carboplatin. The composition can comprise an anti-inflammatory agent. The anti-CD20 polypeptide antibodies can be humanized antibodies. The anti-CD20 polypeptide antibodies can be chimeric antibodies. The composition can be administered by injection. The administration of the composition can be effective to increase progression-free survival by 25 percent. The administration of the composition can be effective to increase progression-free survival by 50 percent. The administration of the composition can be effective to increase progression-free survival by 75 percent. The administration of the composition can be effective to increase progression-free survival by 100 percent. The administration of the composition can be under conditions wherein the median time to progression for a population of mammals with the lymphoma is at least 150 days. The administration of the composition can be under conditions wherein the median time to progression for a population of mammals with the lymphoma is at least 165 days. The administration of the composition can be under conditions wherein the median time to progression for a population of mammals with the lymphoma is at least 170 days. The average diameter of the complexes can be from 0.1 μm to 0.3 μm . The average diameter of the complexes can be from 0.15 μm to 0.3 μm . The average diameter of the complexes can be from 0.2 μm to 0.5 μm . The average diameter of the complexes can be from 0.3 μm to 0.5 μm . The average diameter of the complexes can be from 0.2 μm to 0.8 μm . The average diameter of the complexes can be from 0.2 μm to 0.7 μm .

(9) In another aspect, this document features a method for treating a mammal having lymphoma. The method comprises, or consists essentially of, administering, to the mammal, a composition comprising albumin-containing nanoparticle/antibody complexes, wherein the average diameter of at least 5 percent of the complexes of the composition is between 0.1 and 0.9 μm , and wherein the antibodies are anti-CD20 antibodies. The mammal can be a human. The lymphoma can be a mature B-cell neoplasm. The lymphoma can be a mature T cell neoplasm. The lymphoma can be a Hodgkin lymphoma. The albumin-containing nanoparticle/antibody complexes can be ABRAXANE®/Rituximab complexes. The composition or the albumin-containing nanoparticle/antibody complexes can comprise an alkylating agent. The alkylating agent can be a

platinum compound. The platinum compound can be carboplatin. The composition can comprise an anti-inflammatory agent. The anti-CD20 polypeptide antibodies can be humanized antibodies. The anti-CD20 polypeptide antibodies can be chimeric antibodies. The composition can be administered by injection. The administration of the composition can be effective to increase progression-free survival by 25 percent. The administration of the composition can be effective to increase progression-free survival by 50 percent. The administration of the composition can be effective to increase progression-free survival by 75 percent. The administration of the composition can be effective to increase progression-free survival by 100 percent. The administration of the composition can be under conditions wherein the median time to progression for a population of mammals with the lymphoma is at least 150 days. The administration of the composition can be under conditions wherein the median time to progression for a population of mammals with the lymphoma is at least 165 days. The administration of the composition can be under conditions wherein the median time to progression for a population of mammals with the lymphoma is at least 170 days. The average diameter of at least 5 percent of the complexes of the composition can be from 0.2 μm to 0.9 μm . The average diameter of at least 5 percent of the complexes of the composition can be from 0.2 μm to 0.8 μm . The average diameter of at least 5 percent of the complexes of the composition can be from 0.2 μm to 0.7 μm . The average diameter of at least 5 percent of the complexes of the composition can be from 0.2 μm to 0.6 μm . The average diameter of at least 5 percent of the complexes of the composition can be from 0.2 μm to 0.5 μm . The average diameter of at least 5 percent of the complexes of the composition can be from 0.2 μm to 0.4 μm . The average diameter of at least 10 percent of the complexes of the composition can be between 0.1 and 0.9 μm . The average diameter of at least 50 percent of the complexes of the composition can be between 0.1 and 0.9 μm . The average diameter of at least 75 percent of the complexes of the composition can be between 0.1 and 0.9 μm . The average diameter of at least 90 percent of the complexes of the composition can be between 0.1 and 0.9 μm .

(10) Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention pertains.

Although methods and materials similar or equivalent to those described herein can be used to practice the invention, suitable methods and materials are described below. All publications, patent applications, patents, and other references mentioned herein are incorporated by reference in their entirety. In case of conflict, the present specification, including definitions, will control. In addition, the materials, methods, and examples are illustrative only and not intended to be limiting.

(11) The details of one or more embodiments of the invention are set forth in the accompanying drawings and the description below. Other features, objects, and advantages of the invention will be apparent from the description and drawings, and from the claims.

Description

DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a graph plotting the particle size distribution for ABRAXANE® (ABX) dissolved in RITUXAN® (RIT) as determined using a Mastersizer 2000E (Malvern Instruments Ltd, Worcestershire, England). ABX (10 mg/mL) was reconstituted in 1 mL of the indicated amount of RIT, and the mixtures were incubated at room temperature for 30 minutes.

(2) FIG. 2 is a graph plotting percent change at seven days in tumor size from baseline of lymphoma (Daudi cell line) tumor bearing nude mice treated with PBS, RITUXAN® (RIT; 12 mg/kg) only, ABRAXANE® (30 mg/kg) only, RITUXAN® (RIT; 12 mg/kg) followed by ABRAXANE® (30 mg/kg) one day later, or AR160 complexes (0.159 μm ; complexes where ABRAXANE® (10 mg/mL) was premixed with 2 mg/mL RITUXAN® and incubated for 60 minutes before injection).

(3) FIG. 3 is a Kaplan Meier graph plotting survival of lymphoma (Daudi cell line) tumor bearing nude mice treated with PBS, RITUXAN® (RIT; 12 mg/kg) only, ABRAXANE® (30 mg/kg) only, RITUXAN® (RIT; 12 mg/kg) followed by ABRAXANE® (30 mg/kg) one day later, or AR160 complexes (0.159 μm ; complexes where ABRAXANE® (10 mg/mL) was premixed with 2 mg/mL RITUXAN® and incubated for 60 minutes before injection).

DETAILED DESCRIPTION

(4) This document provides methods and materials involved in treating lymphomas (e.g., mature B-cell neoplasms, mature T cell neoplasms, mature natural killer cell neoplasms, immunodeficiency-associated lymphoproliferative disorders, Hodgkin lymphomas, and non-Hodgkin lymphomas). For example, this document provides methods and materials for using complexes containing albumin-containing nanoparticles (e.g., ABRAXANE® nanoparticles) and antibodies (e.g., anti-CD20 polypeptide antibodies such as Rituximab) to treat lymphomas.

(5) The methods and materials provided herein can be used to treat any type of lymphoma. For example, the methods and materials provided herein can be used to treat mature B-cell neoplasms, mature T cell neoplasms, mature natural killer cell neoplasms, immunodeficiency-associated lymphoproliferative disorders, Hodgkin lymphomas, or non-Hodgkin lymphomas. In some cases, the methods and materials provided herein can be used to treat lymphomas in any type of mammal including, without limitation, mice, rats, dogs, cats, horses, cows, pigs, monkeys, and humans.

(6) In some cases, complexes containing albumin-containing nanoparticles (e.g., ABRAXANE® nanoparticles) and antibodies (e.g., anti-CD20 polypeptide antibodies such as Rituximab) can be designed to have an average diameter that is greater than 1 μm . For example, appropriate concentrations of albumin-containing nanoparticles and antibodies can be used such that complexes having an average diameter that is greater than 1 μm are formed. In some cases, manipulations such as centrifugation can be used to form preparations of albumin-containing nanoparticle/antibody complexes where the average diameter of those complexes is greater than 1 μm . In some cases, the preparations of albumin-containing nanoparticle/antibody complexes provided herein can have an average diameter that is between 1 μm and 5 μm (e.g., between 1.1 μm and 5 μm , between 1.5 μm and 5 μm , between 2 μm and 5 μm , between 2.5 μm and 5 μm , between 3 μm and 5 μm , between 3.5 μm and 5 μm , between 4 μm and 5 μm , between 4.5 μm and 5 μm , between 1.1 μm and 4.5 μm , between 1.1 μm and 4 μm , between 1.1 μm and 3.5 μm , between 1.1 μm and 3 μm , between 1.1 μm and 2.5 μm , between 1.1 μm and 2 μm , or between 1.1 μm and 1.5 μm). Preparations of albumin-containing nanoparticle/antibody complexes provided herein having an average diameter that is between 1 μm and 5 μm can be administered systemically (e.g., intravenously) to treat lymphomas located within a mammal's body. In some cases, the preparations of albumin-containing nanoparticle/antibody complexes provided herein can have an average diameter that is between 5 μm and 50 μm (e.g., between 6 μm and 50 μm , between 7 μm and 50 μm , between 10 μm and 50 μm , between 15 μm and 50 μm , between 20 μm and 50 μm , between 25 μm and 50 μm , between 30 μm and 50 μm , between 35 μm and 50 μm , between 5 μm and 45 μm , between 5 μm , and 40 μm , between 5 μm and 35 μm , between 5 μm and 30 μm , between 5 μm and 25 μm , between 5 μm and 20 μm , between 5 μm and 15 μm , or between 10 μm and 30 μm). Preparations of albumin-containing nanoparticle/antibody complexes provided herein having an average diameter that is between 5 μm and 50 μm can be administered into a tumor (e.g., intratumorally) or in a region of a tumor located within a mammal's body.

(7) In some cases, a preparation of albumin-containing nanoparticle/antibody complexes provided herein can have greater than 60 percent (e.g., greater than 65, 70, 75, 80, 90, 95, or 99 percent) of the complexes having a diameter that is between 1 μm and 5 μm (e.g., between 1.1 μm and 5 μm , between 1.5 μm and 5 μm , between 2 μm and 5 μm , between 2.5 μm and 5 μm , between 3 μm and 5 μm , between 3.5 μm and 5 μm , between 4 μm and 5 μm , between 4.5 μm and 5 μm , between 1.1 μm and 4.5 μm , between 1.1 μm and 4 μm , between 1.1 μm and 3.5 μm , between 1.1 μm and 3 μm , between 1.1 μm and 2.5 μm , between 1.1 μm and 2 μm , or between 1.1 μm and 1.5 μm).

Preparation of albumin-containing nanoparticle/antibody complexes provided herein having greater than 60 percent (e.g., greater than 65, 70, 75, 80, 90, 95, or 99 percent) of the complexes with a diameter that is between 1 μm and 5 μm can be administered systemically (e.g., intravenously) to treat lymphomas located within a mammal's body. In some cases, a preparation of albumin-containing nanoparticle/antibody complexes provided herein can have greater than 60 percent (e.g., greater than 65, 70, 75, 80, 90, 95, or 99 percent) of the complexes having a diameter that is between 5 μm and 50 μm (e.g., between 6 μm and 50 μm , between 7 μm and 50 μm , between 10 μm and 50 μm , between 15 μm and 50 μm , between 20 μm and 50 μm , between 25 μm and 50 μm , between 30 μm and 50 μm , between 35 μm and 50 μm , between 5 μm and 45 μm , between 5 μm and 40 μm , between 5 μm and 35 μm , between 5 μm and 30 μm , between 5 μm and 25 μm , between 5 μm and 20 μm , between 5 μm and 15 μm , or between 10 μm and 30 μm). Preparation of albumin-containing nanoparticle/antibody complexes provided herein having greater than 60 percent (e.g., greater than 65, 70, 75, 80, 90, 95, or 99 percent) of the complexes with a diameter that is between 5 μm and 50 μm can be administered into a tumor (e.g., intratumorally) or in a region of a tumor located within a mammal's body.

(8) In some cases, complexes containing albumin-containing nanoparticles (e.g., ABRAXANE® nanoparticles) and antibodies (e.g., anti-CD20 polypeptide antibodies such as Rituximab) can be designed to have an average diameter that is less than 1 μm . For example, appropriate concentrations of albumin-containing nanoparticles and antibodies (e.g., Rituximab) can be used such that complexes having an average diameter that is less than 1 μm are formed. In some cases, the preparations of albumin-containing nanoparticle/antibody complexes provided herein can have an average diameter that is between 0.1 μm and 1 μm (e.g., between 0.1 μm and 0.95 μm , between 0.1 μm and 0.9 μm , between 0.1 μm and 0.8 μm , between 0.1 μm and 0.7 μm , between 0.1 μm and 0.6 μm , between 0.1 μm and 0.5 μm , between 0.1 μm and 0.4 μm , between 0.1 μm and 0.3 μm , between 0.1 μm and 0.2 μm , between 0.2 μm and 1 μm , between 0.3 μm and 1 μm , between 0.4 μm and 1 μm , between 0.5 μm and 1 μm , between 0.2 μm and 0.6 μm , between 0.3 μm and 0.6 μm , between 0.2 μm and 0.5 μm , or between 0.3 μm and 0.5 μm). Preparations of albumin-containing nanoparticle/antibody complexes provided herein having an average diameter that is between 0.1 μm and 0.9 μm can be administered systemically (e.g., intravenously) to treat lymphomas located within a mammal's body.

(9) In some cases, a preparation of albumin-containing nanoparticle/antibody complexes provided herein can have greater than 60 percent greater than 65, 70, 75, 80, 90, 95, or 99 percent) of the complexes having a diameter that is between 0.1 μm and 0.9 μm (e.g., between 0.1 μm and 0.95 μm , between 0.1 μm and 0.9 μm , between 0.1 μm and 0.8 μm , between 0.1 μm and 0.7 μm , between 0.1 μm and 0.6 μm , between 0.1 μm and 0.5 μm , between 0.1 μm and 0.4 μm , between 0.1 μm and 0.3 μm , between 0.1 μm and 0.2 μm , between 0.2 μm and 1 μm , between 0.3 μm and 1 μm , between 0.4 μm and 1 μm , between 0.5 μm and 1 μm , between 0.2 μm and 0.6 μm , between 0.3 μm and 0.6 μm , between 0.2 μm and 0.5 μm , or between 0.3 μm and 0.5 μm). Preparation of albumin-containing nanoparticle/antibody complexes provided herein having greater than 60 percent (e.g., greater than 65, 70, 75, 80, 90, 95, or 99 percent) of the complexes with a diameter that is between 0.1 μm and 0.9 μm can be administered systemically (e.g., intravenously) to treat cancers located within a mammal's body.

(10) In general, albumin-containing nanoparticles such as ABRAXANE® can be contacted with an antibody such as an anti-CD20 polypeptide antibody (e.g., Rituximab) prior to administration to a human to form an albumin-containing nanoparticle/antibody complex (e.g., an ABRAXANE®/anti-CD20 polypeptide antibody complex). Any appropriate albumin-containing nanoparticle preparation and any appropriate antibody can be used as described herein. For example, ABRAXANE® nanoparticles can be used as described herein. Examples of antibodies that can be used to form albumin-containing nanoparticle/antibody complexes as described herein include, without limitation, Rituximab (e.g. RITUXAN®, MABTHERA™, or ZYTUX™). For

example, an appropriate dose of ABRAXANE® and an appropriate dose of Rituximab can be mixed together in the same container. This mixture can be incubated at an appropriate temperature (e.g., room temperature, between 15° C. and 30° C., between 15° C. and 25° C., between 20° C. and 30° C., or between 20° C. and 25° C.) for a period of time (e.g., about 30 minutes, or between about 5 minutes and about 60 minutes, between about 5 minutes and about 45 minutes, between about 15 minutes and about 60 minutes, between about 15 minutes and about 45 minutes, between about 20 minutes and about 400 minutes. or between about 25 minutes and about 35 minutes) before being administered to a cancer patient (e.g., a lymphoma patient). In some cases, ABRAXANE® can be contacted with an anti-CD20 polypeptide antibody by injecting both ABRAXANE® and the anti-CD20 polypeptide antibody either individually or as a pre-mixed combination into an IV bag containing an IV bag solution. The contents of the IV bag including ABRAXANE®/anti-CD20 polypeptide antibody complexes can be introduced into the patient to be treated.

(11) In some cases, albumin-containing nanoparticles such as ABRAXANE® can be contacted with an antibody such as an anti-CD20 polypeptide antibody (e.g., Rituximab) to form albumin-containing nanoparticle/antibody complexes (e.g., ABRAXANE®/anti-CD20 polypeptide antibody complexes) that are stored prior to being administered to a cancer patient (e.g., a lymphoma patient). For example, a composition containing albumin-containing nanoparticle/antibody complexes can be formed as described herein and stored for a period of time (e.g., days or weeks) prior to being administered to a cancer patient.

(12) Any appropriate method can be used to obtain albumin-containing nanoparticles such as ABRAXANE® and an antibody such as an anti-CD20 polypeptide antibody. For example, ABRAXANE® can be obtained from Celgene Corp. or as described elsewhere (U.S. Pat. No. 6,537,579). Rituximab can be obtained from Genentech Corp. or Roche Corp. or as described elsewhere (U.S. Pat. No. 5,736,137).

(13) In some cases, the combination of an albumin-containing nanoparticle such as ABRAXANE® and an antibody such as anti-CD20 polypeptide antibody can include one or more other agents such as an alkylating agent (e.g., a platinum compound). Examples of platinum compounds that can be used as an alkylating agent include, without limitation, carboplatin (PARAPLATIN®), cisplatin (PLATINOL®) oxaliplatin (ELOXATIN®), and BBR3464. Examples of other agents that can be included within an albumin-containing nanoparticle/antibody complex provided herein include, without adriamycin, cyclophosphamide, vincristine, prednisone, dexamethasone, cytarabine, methotrexate, thiotepa, ifosfamide, chlorambucil, dacarbazine, bleomycin, ibrutinib, campath-B, gemcitabine, revlimid, sirolimus, temsirolimus, bexxar, brentuximab, bendamustine, and etoposide. For example, an albumin-containing nanoparticle/antibody complex provided herein (e.g., ABRAXANE®/anti-CD20 polypeptide antibody complex) can include brentuximab, cyclophosphamide, adriamycin, or vincristine as part of the complex.

(14) Any appropriate method can be used to administer an albumin-containing nanoparticle/antibody complex provided herein (e.g., ABRAXANE®/anti-CD20 polypeptide antibody complexes) to a mammal. For example, a composition containing albumin-containing nanoparticle/antibody complexes such as ABRAXANE®/anti-CD20 polypeptide antibody complexes can be administered via injection (e.g., subcutaneous injection, intramuscular injection, intravenous injection, or intrathecal injection).

(15) Before administering a composition containing an albumin-containing nanoparticle/antibody complex provided herein (e.g., ABRAXANE®/anti-CD20 polypeptide antibody complexes) to a mammal, the mammal can be assessed to determine whether or not the mammal has lymphoma. Any appropriate method can be used to determine whether or not a mammal has lymphoma. For example, a mammal (e.g., human) can be identified as having lymphoma using standard diagnostic techniques. In some cases, a tissue biopsy (e.g., lymph node tissue sample) can be collected and analyzed to determine whether or not a mammal has lymphoma.

(16) After identifying a mammal as having lymphoma, the mammal can be administered a composition containing albumin-containing nanoparticle/antibody complexes provided herein (e.g., ABRAXANE®/anti-CD20 polypeptide antibody complexes). For example, a composition containing ABRAXANE®/anti-CD20 polypeptide antibody complexes can be administered prior to or in lieu of surgical resection of a tumor. In some cases, a composition containing albumin-containing nanoparticle/antibody complexes provided herein (e.g., ABRAXANE®/anti-CD20 polypeptide antibody complexes) can be administered following resection of a tumor.

(17) A composition containing albumin-containing nanoparticle/antibody complexes provided herein (e.g., ABRAXANE®/anti-CD20 polypeptide antibody complexes) can be administered to a mammal in any appropriate amount, at any appropriate frequency, and for any appropriate duration effective to achieve a desired outcome (e.g., to increase progression-free survival). In some cases, a composition containing albumin-containing nanoparticle/antibody complexes provided herein (e.g., ABRAXANE®/anti-CD20 polypeptide antibody complexes) can be administered to a mammal having lymphoma to reduce the progression rate of the lymphoma by 5, 10, 25, 50, 75, 100, or more percent. For example, the progression rate can be reduced such that no additional cancer progression is detected. Any appropriate method can be used to determine whether or not the progression rate of lymphoma is reduced. For example, the progression rate of lymphoma can be assessed by imaging tissue at different time points and determining the amount of cancer cells present. The amounts of cancer cells determined within tissue at different times can be compared to determine the progression rate. After treatment as described herein, the progression rate can be determined again over another time interval. In some cases, the stage of cancer (e.g., lymphoma) after treatment can be determined and compared to the stage before treatment to determine whether or not the progression rate was reduced.

(18) In some cases, a composition containing albumin-containing nanoparticle/antibody complexes provided herein (e.g., ABRAXANE®/anti-CD20 polypeptide antibody complexes) can be administered to a mammal having lymphoma under conditions where progression-free survival is increased (e.g., by 5, 10, 25, 50, 75, 100, or more percent) as compared to the median progression-free survival of corresponding mammals having untreated lymphoma or the median progression-free survival of corresponding mammals having lymphoma treated with ABRAXANE® and an antibody (e.g., an anti-CD20 polypeptide antibody) without forming ABRAXANE®/antibody complexes (e.g., without forming ABRAXANE®/anti-CD20 polypeptide antibody complexes). In some cases, a composition containing albumin-containing nanoparticle/antibody complexes provided herein (e.g., ABRAXANE®/anti-CD20 polypeptide antibody complexes) can be administered to a mammal having lymphoma to increase progression-free survival by 5, 10, 25, 50, 75, 100, or more percent as compared to the median progression-free survival of corresponding mammals having lymphoma and having received ABRAXANE® or an antibody (e.g., an anti-CD20 polypeptide antibody) alone. Progression-free survival can be measured over any length of time (e.g., one month, two months, three months, four months, five months, six months, or longer).

(19) In some cases, a composition containing albumin-containing nanoparticle/antibody complexes provided herein (e.g., ABRAXANE®/anti-CD20 polypeptide antibody complexes) can be administered to a mammal having lymphoma under conditions where the 8-week progression-free survival rate for a population of mammals is 65% or greater (e.g., 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80% or greater) than that observed in a population of comparable mammals not receiving a composition containing albumin-containing nanoparticle/antibody complexes provided herein (e.g., ABRAXANE®/anti-CD20 polypeptide antibody complexes). In some cases, a composition containing albumin-containing nanoparticle/antibody complexes provided herein (e.g., ABRAXANE®/anti-CD20 polypeptide antibody complexes) can be administered to a mammal having lymphoma under conditions where the median time to progression for a population of mammals is at least 150 days (e.g., at least 155, 160, 163, 165, or 170 days).

(20) An effective amount of a composition containing albumin-containing nanoparticle/antibody complexes provided herein e.g., ABRAXANE®/anti-CD20 polypeptide antibody complexes) can be any amount that reduces the progression rate of lymphoma, increases the progression-free survival rate, or increases the median time to progression without producing significant toxicity to the mammal. Typically, an effective amount of ABRAXANE® can be from about 50 mg/m² to about 150 mg/m² (e.g., about 80 mg/m²), and an effective amount of an anti-CD20 polypeptide antibody such as Rituximab can be from about 5 mg/kg to about 20 mg/kg (e.g., about 10 mg/kg or 375 mg/m²). If a particular mammal fails to respond to a particular amount, then the amount of ABRAXANE® or anti-CD20 polypeptide antibody can be increased by, for example, two fold. After receiving this higher concentration, the mammal can be monitored for both responsiveness to the treatment and toxicity symptoms, and adjustments made accordingly. The effective amount can remain constant or can be adjusted as a sliding scale or variable dose depending on the mammal's response to treatment. Various factors can influence the actual effective amount used for a particular application. For example, the frequency of administration, duration of treatment, use of multiple treatment agents, route of administration, and severity of the lymphoma may require an increase or decrease in the actual effective amount administered.

(21) The frequency of administration can be any frequency that reduces the progression rate of lymphoma, increases the progression-free survival rate, or increases the median time to progression without producing significant toxicity to the mammal. For example, the frequency of administration can be from about once a month to about three times a month, or from about twice a month to about six times a month, or from about once every two months to about three times every two months. The frequency of administration can remain constant or can be variable during the duration of treatment. A course of treatment with a composition containing ABRAXANE®/anti-CD20 polypeptide antibody complexes can include rest periods. For example, a composition containing ABRAXANE®/anti-CD20 polypeptide antibody complexes can be administered over a two week period followed by a two week rest period, and such a regimen can be repeated multiple times. As with the effective amount, various factors can influence the actual frequency of administration used for a particular application. For example, the effective amount, duration of treatment, use of multiple treatment agents, route of administration, and severity of the lymphoma may require an increase or decrease in administration frequency.

(22) An effective duration for administering a composition provided herein can be any duration that reduces the progression rate of lymphoma, increases the progression-free survival rate, or increases the median time to progression without producing significant toxicity to the mammal. Thus, the effective duration can vary from several days to several weeks, months, or years. In general, the effective duration for the treatment of lymphoma can range in duration from several weeks to several months. In some cases, an effective duration can be for as long as an individual mammal is alive. Multiple factors can influence the actual effective duration used for a particular treatment. For example, an effective duration can vary with the frequency of administration, effective amount, use of multiple treatment agents, route of administration, and severity of the lymphoma.

(23) A composition containing albumin-containing nanoparticle/antibody complexes provided herein (e.g., ABRAXANE®/anti-CD20 polypeptide antibody complexes) can be in any appropriate form. For example, a composition provided herein can be in the form of a solution or powder with or without a diluent to make an injectable suspension. A composition also can contain additional ingredients including, without limitation, pharmaceutically acceptable vehicles. A pharmaceutically acceptable vehicle can be, for example, saline, water, lactic acid, mannitol, or combinations thereof.

(24) After administering a composition provided herein to a mammal, the mammal can be monitored to determine whether or not the lymphoma was treated. For example, a mammal can be assessed after treatment to determine whether or not the progression rate of lymphoma was reduced (e.g., stopped). As described herein, any method can be used to assess progression and survival

rates.

(25) In some cases, a formulation of ABRAXANE®/RITUXAN® complexes described in Example 1 can be administered to a human lymphoma patient as described in the methods set forth in Example 3.

(26) In some cases, nanoparticles containing albumin (e.g., nanoparticles with an albumin shell) and an agent other than paclitaxel can be used as described herein in place of or in combination with ABRAXANE®. For example, albumin-containing nanoparticles designed to carry a cancer chemotherapeutic agent can be used to form nanoparticle/anti-CD20 polypeptide antibody complexes that can be used as described herein. An example of such a cancer chemotherapeutic agent includes, without limitation, vinblastine.

(27) In some cases, a composition can be formulated to include nanoparticles containing albumin (e.g., nanoparticles with an albumin shell) that are conjugated to an antibody, agent, or combination of antibodies and agents to form complexes for treating lymphoma. For example, albumin nanoparticles can be formulated to include adriamycin, cyclophosphamide, vincristine, prednisone, dexamethasone, cytarabine, methotrexate, thiotepa, ifosfamide chlorambucil, dacarbazine, bleomycin, ibrutinib, campath-B, gemcitabine, revlimid, sirolimus temsirolimus, bexxar, brentuximab, bendamustine, etoposide, or combinations thereof with or without including rituximab.

(28) In some cases, nanoparticles containing albumin (e.g., nanoparticles with an albumin shell) or a complex described herein (e.g., ABRAXANE®/rituximab complexes) can be formulated to include one or more anti-chronic inflammation treatment agents designed to reduce the global state of immune dysfunction and/or chronic inflammation present within a cancer patient. For example, steroidal anti-inflammatory agents (e.g., prednisone), non-steroidal anti-inflammatory agents (e.g., naproxen), lympho-depleting cytotoxic agents (e.g., cyclophosphamide), immune cell and/or cytokine targeting antibodies (e.g., infliximab), or a combination thereof can be incorporated into nanoparticles containing albumin or ABRAXANE®/rituximab complexes. In some cases, anti-IL-4 agents (e.g., anti-IL-4 antibodies), anti-IL-13 agents (e.g., soluble IL-13 receptor), and combinations thereof can be incorporated into nanoparticles containing albumin or ABRAXANE®/rituximab complexes.

(29) Any appropriate method can be used to assess whether or not the global state of immune dysfunction and/or chronic inflammation was reduced following an anti-chronic inflammation treatment. For example, cytokine profiles (e.g., IL-4, IL-13, IL-4, IL-13, IL-5, IL-10, IL-2, and interferon gamma) present in blood can be assessed before and after an anti-chronic inflammation treatment to determine whether or not the global state of immune dysfunction and/or chronic inflammation was reduced.

(30) The invention will be further described in the following examples, which do not limit the scope of the invention described in the claims.

EXAMPLES

Example 1—Making ABRAXANE®/RITUXAN® Complexes

(31) ABRAXANE® was incubated with various increasing concentrations of RITUXAN® (rituximab) to form ABRAXANE®/RITUXAN® complexes of increasing diameter. Ten milligrams of ABRAXANE® was reconstituted in 1 mL of RITUXAN® 0, 2, 4, 6, 8, and 10 mg/mL, and the mixture was incubated at room temperature for 30 minutes. After incubation, the distributions of particle sizes were determined with the Mastersizer 2000. The median particle size ranged from 0.147 μm to 8.286 μm for 0 and 10 mg/mL RITUXAN®, respectively (FIG. 1). These results demonstrate that the antibody concentration in which ABRAXANE® is incubated impacts the size of the nanoparticle. Manipulating the size of the particles can change the pharmacokinetics of the drug complex as well as its bio-distribution, which in turn can improve the clinical efficacy of the drug complex.

Example 2—ABRAXANE®/RITUXAN® Complexes Inhibit Tumor Growth More Effectively

than ABRAXANE® Alone, RITUXAN® Alone, and the Sequential Use of RITUXAN® and-ABRAXANE®

(32) Female athymic nude mice were injected with 1×10^6 lymphoma cells (Daudi cell line). Tumors were allowed to grow, and treatments were administered when tumors were between 800 and 1000 mm³. Mice were treated with (a) a single dose of 100 μ L PBS on day 0, (b) a single dose of RITUXAN® (12 mg/kg) on day 0, (c) a single dose of ABRAXANE® (30 mg/kg) on day 0, (d) a single dose of RITUXAN® (12 mg/kg) on day 0 followed by a single dose of ABRAXANE® (30 mg/kg) on day 1, or (e) a single dose of AR160 (equivalent to 30 mg/kg of ABRAXANE®) on day 0. Tumor volumes (mm³) were monitored, and the percent change in tumor volume seven days after treatment was determined.

(33) The AR 160 complexes were produced as follows. 10 mg ABRAXANE® was reconstituted in 2 mg of RITUXAN® in 500 μ L 0.9% saline and incubated for 1 hour at room temperature. After incubation, AR160 was brought to 1 mL with 0.9% saline. AR160 was further diluted and 100 μ L was administered to mice at a dose equivalent to 12 mg/kg RITUXAN® and 30 mg/kg ABRAXANE®. Average particle size for AR160 was 0.159 μ m.

(34) On day 7 post treatment, the mice treated with AR160 exhibited tumors with significantly smaller tumor size as compared to the other treatment groups (FIG. 2). Survival data also revealed an improvement for mice treated with ABRAXANE®/RITUXAN® complexes (FIG. 3). These results demonstrate that the albumin-containing nanoparticle/antibody complexes provided herein e.g., ABRAXANE®/anti-CD20 polypeptide antibody complexes) can be used effectively to treat lymphomas.

Example 3—ABRAXANE®/RITUXAN® Complexes as Targeted Therapy for Lymphomas

(35) The treatment schedule for ABRAXANE®/RITUXAN® complexes is repeated each month (every 28 days $\pm$ 3 days) or until disease progression, patient refusal, or unacceptable toxicity (Table 1) with the indicated dose escalation scheme (Table 2) and dose limiting toxicities (Table 3).

(36) TABLE-US-00001 TABLE 1 Agent Dose Route Days ReRx ABRAXANE®/ assigned IV over 60 minutes 1, 8 Every Rituxan at time of (only 1st dose; and 15 28 days* complexes registration subsequent doses infused over 30 minutes) *One treatment cycle = 28 days $\pm$ 3 days

(37) TABLE-US-00002 TABLE 2 Dose Escalation Scheme. Dose Level Dose (ABX) Dose (RIT) -2 75 mg/m² 30 mg/m² -1 100 mg/m² 40 mg/m² 1* 125 mg/m² 50 mg/m² 2 150 mg/m² 60 mg/m² 3 175 mg/m² 70 mg/m² *Starting dose.

(38) TABLE-US-00003 TABLE 3 Dose Limiting Toxicities (DLT). Toxicity DLT Definition Hematologic Grade 4 ANC, Grade 4 Hgb, or PLT < 25,000 Renal Serum creatinine $\geq$ 2 times baseline Other $\geq$ grade 3 as per NCI Common Terminology nonhematologic Criteria for Adverse Events (CTCAE) version 4.0

Determination of Maximum Tolerated Dose (MTD)

(39) The maximum tolerated dose is defined as the highest dose level among those tested where at most one out of six patients develops a DLT prior to the start of their second cycle of treatment and the next highest dose level is such that two out of a maximum of six patients treated at this dose level developed a DLT prior to the start of their second cycle of treatment.

(40) Enrollment and Determination of MTD

(41) A minimum of two or a maximum of six patients are accrued to a given dose level. For dose level 1 (and if accrued to, dose levels -1 & -2), enrollment is temporarily halted after each patient has been enrolled in order to gather acute adverse event data over the first cycle of their treatment. For dose levels 2 & 3, patients are accrued to these dose levels so that at any given time no more than two patients are receiving their first cycle of treatment and acute adverse event data over the first treatment cycle for all other patients treated at the current dose level is known. If, at any time in the enrollment process, two patients treated at the current dose level develop a DLT during the first cycle of treatment, enrollment is closed to that dose level. Enrollment is re-opened to the next lower dose level if fewer than six patients have been treated at that dose level. If none of the first

three patients treated at a given dose level develops a DLT during the first cycle of treatment, enrollment to the dose level is closed and enrollment is reopened at next higher dose level. If there are no other higher dose levels to be tested, three additional patients are enrolled at the current dose level to confirm MTD. If one of the first three patients treated at a given dose level develops a DLT during the first cycle of treatment, three additional patients are enrolled (sequentially) onto the current dose level. If, at any time in the enrollment of these three additional patients, a patient develops a DLT, enrollment is closed to this dose level. Enrollment is re-opened to the next lower dose level if fewer than six patients are treated at that dose level. If none of these three additional patients develops a DLT during the first cycle of treatment, enrollment to this dose level is closed and enrollment is reopened at next higher dose level. If there are no other higher dose levels to be tested, this is considered the MTD.

(42) For this protocol, the patient returns for evaluation and retreatment (at least every 28+/-3 days) according to the schedule. If a patient fails to complete the first cycle of treatment for reasons other than toxicity, an additional patient is enrolled to replace this patient.

(43) Dosage Modification Based on Adverse Events

(44) The modifications in Table 4 are followed until individual treatment tolerance is ascertained. If multiple adverse events (Table 5) are seen, dose is administered based on greatest reduction required for any single adverse event observed. Dose modifications apply to the treatment given in the preceding cycle and are based on adverse events observed since the prior dose.

(45) TABLE-US-00004 TABLE 4 Dose Levels Based on Adverse Events. ABRAXANE

®/RITUXAN ® complexes - Both drugs are reduced Dose Accompanying RIT dose Level ABX dose (40% of ABX dose) 2 175 mg/m² 70 mg/m² -1 150 mg/m² 60 mg/m² 1 125 mg/m² 50 mg/m² -2 100 mg/m² 40 mg/m² -2 75 mg/m² 30 mg/m² *Dose level 1 refers to the starting dose.

(46) TABLE-US-00005 TABLE 5 Use Common Terminology Criteria for Adverse Events

(CTCAE) v. 4.0* unless otherwise specified CTCAE Category Adverse Event Dose Reduction Investigations ANC < 1000 or Day 1: Hold until counts above these levels. PLT < 75,000 Day 8: Omit dose that day and retreat at same dose level on day 15 if counts have recovered. Day 15: Omit dose that day. NOTE: if two consecutive cycles of therapy require omission of a dose, subsequent treatment cycles should begin (day 1) at next lower dose. AST or Alkaline Day 1: Hold until resolved to < Grade 2 then Phosphatase reduce dose by ONE dose level. ≥Grade 2 If treatment needs to be held >4 weeks, discontinue study treatment and go to event monitoring. Neurology Neuropathy Day 1: Hold until resolved to < Grade 2 then disorders ≥Grade 2 reduce dose by ONE dose level. Day 8 OR 15—Omit dose that day. If resolved to < Grade 2 by next scheduled dose, then dose reduce by one level If treatment needs to be held >4 weeks, discontinue study treatment and go to Event Monitoring All other non- ≥Grade 3 Day 1: Hold until resolved to ≤ Grade 2 then hematologic reduce dose by ONE dose level. adverse events Day 8: Omit dose that day. If resolved to ≤ Grade 2 by day 15, then dose reduce by one level and retreat. Day 15: Omit dose that day. NOTE: if two consecutive cycles of therapy require omission of a dose, subsequent treatment cycles should begin (day 1) at next lower dose. If treatment needs to be held >4 weeks, discontinue study treatment and go to Event Monitoring Gastrointestinal Bowel perforation Discontinue all study treatment and proceed to Disorders Event Monitoring Bowel Obstruction Grade 1 Continue patient on study for partial bowel obstruction NOT requiring medical intervention. Grade 2 Hold for partial obstruction requiring medical intervention. If resolved to Grade 0 within 4 weeks, treatment may be restarted. If treatment needs to be held >4 weeks, discontinue all study treatment and go to Event Monitoring Grade 3 or 4 For complete bowel obstruction, discontinue study treatment and proceed to Event Monitoring Cardiac Hypertension Hypertension should be treated as per general Disorders ≥Grade 3 practice. If hypertension (≥150/100) persists despite treatment, hold treatment until blood pressure is below this level If treatment needs to be held >4 weeks due to uncontrolled hypertension, discontinue study treatment and go to Event Monitoring. Left

ventricular systolic function- Grade 3 Hold until resolution to Grade ≤ 1 . If treatment needs to be held >4 weeks, discontinue all study treatment and go to Event Monitoring. Grade 4 Discontinue treatment and proceed to Event Monitoring Respiratory, Bronchopulmonary thoracic and Hemorrhage mediastinal $\geq$ Grade 2 Discontinue all study treatment and proceed to disorders Event Monitoring Coagulation Hemorrhage Grade 3 Hold until ALL of the following criteria are met: 1. Bleeding has resolved and Hb is stable. 2. There is no bleeding diathesis that would increase the risk of therapy. 3. There is no anatomic or pathologic condition that could increase the risk of hemorrhage recurrence. If treatment needs to be held >4 weeks, discontinue study treatment and go to Event Monitoring Patients who experience a recurrence of Grade 3 hemorrhage are to discontinue all study treatment and proceed to Event Monitoring. Grade 4 Discontinue study treatment and proceed to Event Monitoring Bleeding diathesis Discontinue study treatment and proceed to Grade 3 or 4 Event Monitoring Vascular Venous thrombosis disorders Grade 3 or Hold treatment. If the planned duration of full- asymptomatic dose anticoagulation is <2weeks, treatment Grade 4 should be held until the full-dose anticoagulation period is over. If the planned duration of full-dose anticoagulation is >2 weeks, treatment may be resumed during the period of full-dose anticoagulation IF all of the criteria below are met: The subject must have an in-range INR (usually 2-3) on a stable dose of warfarin, or on stable dose of heparin prior to restarting treatment. The subject must not have pathological conditions that carry high risk of bleeding (e.g. tumor involving major vessels or other conditions) The subject must not have had hemorrhagic events while on study Symptomatic If thromboemboli worsen/recur upon Grade 4 resumption of study therapy, discontinue treatment. Discontinue treatment and proceed to Event Monitoring Arterial thrombosis Discontinue treatment and proceed to Event (Angina, myocardial Monitoring infarction, transient ischemic attack, cerebrovascular accident, or any other arterial thromboembolic events) ANY Grade

Ancillary Treatment/Supportive Care

(47) Routine use of colony-stimulating factors (G-CSF or GM-CSF) is not recommended.

Prophylactic use of colony-stimulating factors during the study is not allowed. Therapeutic use in patients with serious neutropenic complications such as tissue infection, sepsis syndrome, fungal infection, etc., may be considered at physician discretion. Recombinant erythropoietin to maintain adequate hemoglobin levels and avoid packed red blood cell transfusions is allowed.

(48) Patients should receive full supportive care while on this study. This includes blood product support, antibiotic treatment and treatment of other newly diagnosed or concurrent medical conditions. All blood products and concomitant medications such as antidiarrheals, analgesics, and anti-emetics received from the first administration of study drugs until 30 days after the final dose are to be recorded in the medical record. Patients participating in phase I program clinical trials are not to be considered for enrollment in any other study involving a pharmacologic agent-(drugs, biologics, immunotherapy approaches, gene therapy) whether for symptom control or therapeutic intent.

(49) Hypersensitivity Reactions

(50) Patients do not require premedication prior to administration of ABRAXANE®/RITUXAN® complexes. In the unlikely event of a hypersensitivity reaction, treatment with antihistamines, H2 blockers, and corticosteroids is recommended. Patients should be pre-medicated with the typical regimen for paclitaxel regimens for subsequent cycles. In the unlikely event of a mild hypersensitivity reaction, premedication may be administered using the premedication regimen the institution typically uses for solvent-based paclitaxel.

(51) ABRAXANE®/RITUXAN® Complexes

(52) ABRAXANE®/RITUXAN® complexes are prepared as a hazardous low risk product.

ABRAXANE® is supplied as a white to off-white lyophilized powder containing 100 mg of paclitaxel and approximately 900 mg Albumin Human USP (HA) as a stabilizer in a 50 mL, single-use vial. Each vial of the lyophilized product is reconstituted as set forth below. Unreconstituted

ABRAXANE® is stored at controlled room temperature in its carton. Reconstituted ABRAXANE® is used immediately. RITUXAN® is classified as an anti-CD20 monoclonal antibody.

(53) The dose appropriate number of vials of RITUXAN® are obtained, and each vial is further diluted per the following directions to 4 mg/mL. The dose appropriate number of ABRAXANE® (paclitaxel) 100 mg vials is obtained and each vial is reconstituted per the following directions to a final concentration containing 10 mg/mL nanoparticle albumin-bound (nab) paclitaxel. It is not a requirement to use filter needles in the preparation of, or in-line filters during administration. In addition, filters of pore-size less than 15 micrometers are to be avoided.

(54) As with other cytotoxic anticancer drugs, caution is exercised in handling ABRAXANE®. The use of gloves is recommended.

(55) Using a sterile 3 mL syringe, 1.6 mL (40 mg) of RITUXAN® 25 mg/mL is withdrawn and slowly injected over a minimum of 1 minute onto the inside wall of each of the vials containing 100 mg of ABRAXANE®. Unused RITUXAN® left in the 25 mg/mL vial is discarded, as the product contains no preservatives. Injecting the RITUXAN® solution directly onto the lyophilized cake is avoided as this will result in foaming. Using a sterile 12 mL sterile syringe, 8.4 mL of 0.9% Sodium Chloride Injection, USP, is withdrawn and slowly injected, over a minimum of 1 minute, onto the inside wall of each vial containing ABRAXANE® 100 mg and RITUXAN® 40 mg. Once the addition of RITUXAN® 1.6 mL and 0.9% Sodium Chloride Injection, USP 8.4 mL is complete in each vial, each vial is gently swirled and/or inverted slowly for at least 2 minutes until complete dissolution of any cake/powder occurs. The generation of foam is avoided. The concentration of each vial is 100 mg/10 mL ABRAXANE® and 40 mg/10 mL RITUXAN®. The vials containing the ABRAXANE® and RITUXAN® are allowed to sit for 60 minutes. The vial(s) are gently swirled and/or inverted every 10 minutes to continue to mix the complexes. After 60 minutes is elapsed, a sterile 60- to 100-mL syringe (appropriate size for the volume being administered) is used to withdraw the calculated dosing volume of ABRAXANE® and RITUXAN® from each vial. A sufficient quantity of 0.9% Sodium Chloride Injection, USP is added to make the final concentration of ABRAXANE® 5 mg/mL and RITUXAN® 2 mg/mL. The syringe is gently swirled and/or inverted slowly for 1 minute to mix. The storage and stability is for up to 4 hours at room temperature following final dilution.

(56) Administration

(57) The IV initial complex dose is infused over 60 minutes via syringe pump. The infusion may be shortened to 30 minutes if the initial infusion is well tolerated. Infusion is monitored closely during the infusion process for signs/symptoms of an infusion reaction. The patient's line is flushed after administration with 20 mL 0.9% Sodium Chloride. An example calculation and preparation is as follows: Dose level 1: ABRAXANE® 125 mg/m²/RITUXAN® 50 mg/m². 2 Doses required: ABRAXANE® 250 mg/RITUXAN® 100 mg Obtain three 100 mg vials of ABRAXANE®. Obtain one 100 mg vial of RITUXAN® 25 mg/mL. Withdraw 1.6 mL (40 mg) of RITUXAN® 25 mg/mL and slowly inject over 1 minute onto the inside wall of one of the 100 mg ABRAXANE® vials. Repeat this procedure for each of the remaining two ABRAXANE® 100 mg vials. Add 8.4 mL 0.9% Sodium Chloride Injection, USP onto the inside wall of one of the vials containing ABRAXANE® and RITUXAN®. Repeat this procedure for each of the remaining two ABRAXANE® and RITUXAN® vials. Let mixture sit for 60 minutes (swirling every 10 minutes). The final concentration of each vial should be 100 mg ABRAXANE®/10 mL and 40 mg RITUXAN®/10 mL. Withdraw 25 mL from the ABRAXANE® and RITUXAN® containing vial and place in a 100 mL sterile syringe. Add 25 mL 0.9% Sodium Chloride Injection, USP for a final ABRAXANE® concentration of 5 mg/mL and RITUXAN® concentration of 2 mg/mL. Infuse via syringe pump over 60 minutes (first dose, 30 minutes subsequent doses).

Response to ABRAXANE®/RITUXAN® Complex Treatment

(58) Each patient's response to treatment with an ABRAXANE®/RITUXAN® complex

formulation is monitored.

Other Embodiments

(59) It is to be understood that while the invention has been described in conjunction with the detailed description thereof, the foregoing description is intended to illustrate and not limit the scope of the invention, which is defined by the scope of the appended claims. Other aspects, advantages, and modifications are within the scope of the following claims.

Claims

1. A method for treating a human having lymphoma, said method comprising administering to said human a composition comprising pre-formed nanoparticle/antibody complexes comprising albumin-paclitaxel nanoparticles and an anti-CD20 monoclonal antibody, wherein albumin is complexed to the anti-CD20 monoclonal antibody in the pre-formed nanoparticle/antibody complexes, wherein the nanoparticle/antibody complexes have been pre-formed in vitro by mixing aqueous albumin-paclitaxel nanoparticles with the anti-CD20 monoclonal antibody under conditions to form the nanoparticle/antibody complexes, wherein the conditions comprise incubating at least 5 mg/ml aqueous albumin-paclitaxel nanoparticles with the anti-CD20 monoclonal antibody at a weight ratio of 5:1, 5:2, 5:3, 5:4 or 1:1 albumin-paclitaxel nanoparticle to antibody and for between about 5 minutes and about 400 minutes and diluting the pre-formed nanoparticle/antibody complexes prior to administration, such that the nanoparticle/antibody complexes have CD20 binding specificity and an average diameter of said complexes is between 0.15 μm and 0.9 μm ; said nanoparticle/antibody complexes suitable for administration under conditions wherein a length of progression-free survival is increased, wherein a B-cell or T-cell expresses CD20.
2. The method of claim 1, wherein said lymphoma is a Hodgkin lymphoma.
3. The method of claim 1, wherein the anti-CD20 monoclonal antibody is Rituximab.
4. The method of claim 1, wherein said composition comprises an alkylating agent complexed with said nanoparticles.
5. The method of claim 4, wherein said alkylating agent is a platinum compound.
6. The method of claim 5, wherein said platinum compound is carboplatin.
7. The method of claim 1, wherein said anti-CD20 monoclonal antibody is a humanized antibody or a chimeric antibody.
8. The method of claim 1, wherein said composition is administered by injection.
9. The method of claim 1, wherein said progression-free survival is increased by 15 percent.
10. The method of claim 1, wherein said composition is suitable for administration under conditions wherein a time to progression is increased.
11. A method for treating a human having lymphoma, wherein said method comprises administering, to said human, a composition comprising pre-formed albumin-containing nanoparticle/antibody complexes, wherein the complexes have been pre-formed in vitro by mixing aqueous albumin-paclitaxel nanoparticles with the antibody under conditions to form the complexes, wherein the conditions comprise incubating at least 5 mg/ml aqueous albumin-paclitaxel nanoparticles with the anti-CD20 monoclonal antibody at a weight ratio of 5:1, 5:2, 5:3, 5:4 or 1:1 albumin-paclitaxel nanoparticle to antibody and for between about 5 minutes and about 400 minutes and diluting the pre-formed nanoparticle/antibody complexes prior to administration, wherein albumin is complexed to the antibody in the pre-formed albumin-containing nanoparticle/antibody complexes, wherein an average diameter of said complexes is between 0.15 μm and 0.9 μm , and wherein said antibody is an anti-CD20 monoclonal antibody and the complexes have CD20 binding specificity, wherein a B-cell or T-cell expresses CD20.
12. The method of claim 11, wherein said lymphoma is a mature B-cell neoplasm or a mature T cell neoplasm.

13. The method of claim 11, wherein said composition further comprises an anti-inflammatory agent.

14. The method of claim 11, wherein said administration of said composition is under conditions wherein a median time to progression for a population of humans with said lymphoma is at least 150 days.

15. The method of claim 11, wherein the average diameter of said complexes is from 0.15 μm to 0.3 μm .

16. The method of claim 11, wherein the average diameter of said complexes is from 0.3 μm to 0.5 μm .

17. A method for treating a human having lymphoma, wherein said method comprises administering, to said human, a composition comprising pre-formed albumin-containing nanoparticle/antibody complexes, wherein the complexes have been pre-formed in vitro by mixing aqueous albumin-paclitaxel nanoparticles with an antibody under conditions to form the complexes, wherein the conditions comprise incubating at least 5 mg/ml aqueous albumin-paclitaxel nanoparticles with the anti-CD20 monoclonal antibody at a weight ratio of 5:1, 5:2, 5:3, 5:4 or 1:1 albumin-paclitaxel nanoparticle to antibody and for between about 5 minutes and about 400 minutes and diluting the pre-formed nanoparticle/antibody complexes prior to administration, wherein albumin is complexed to the antibody in the pre-formed albumin-containing nanoparticle/antibody complexes, wherein an average diameter of at least 5 percent of said complexes of said composition is between 0.15 μm and 0.9 μm , and wherein said antibodies are anti-CD20 monoclonal antibodies and the complexes have CD20 binding specificity, wherein a B-cell or T-cell expresses CD20.

18. The method of claim 17, wherein the average diameter of at least 75 percent of said complexes of said composition is between 0.15 μm and 0.9 μm .

19. The method of claim 1, wherein the complexes have been pre-formed in vitro by mixing 5 mg/ml or 10 mg/ml aqueous albumin-paclitaxel nanoparticles with the antibody under conditions to form the complexes.
